GigShield

AI-Powered Parametric Micro-Insurance

for India's Food Delivery Partners

Guidewire DEVTrails 2026

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PERSONA	COVERAGE	PRICING
Zomato & Swiggy Delivery Partners	Income loss from external disruptions only	Weekly subscription Rs. 20 – Rs. 50 / week

Team Member	Role
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Protecting India's delivery backbone, one week at a time.

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1. The Problem

India's food delivery partners earn between **Rs. 15,000–Rs. 25,000 per month**, but **20–30% of that income is at risk** from disruptions they cannot control: monsoon flooding, extreme heat, severe air pollution, curfews, and urban shutdowns.

> **Traditional insurance does not solve this.** It requires manual claims, lengthy verification, and payouts that take days or weeks. By then, the worker has already missed rent, EMI, or school fees. The disruption window is over — the damage is done.

Scale of the problem:

- **15 million+** gig delivery workers in India
- A single Mumbai monsoon day renders **30–40 zones** unworkable for 4–6 hours
- Worker losing 3 hours to flooding loses **Rs. 260–350** with zero recourse
- Less than **2%** of gig workers hold any income protection insurance

Why existing solutions fail:

- **Traditional insurance:** manual claims; weeks to pay; not income-indexed
- **Platform welfare funds:** discretionary; not guaranteed
- **Government schemes (PMSBY/ESIC):** exclude gig workers
- **Personal savings:** workers live week-to-week; no buffer

2. Our Solution — GigShield

GigShield is an **AI-enabled parametric micro-insurance platform** purpose-built for the gig economy. It monitors weather and civic conditions in real time, detects when a worker is disrupted, verifies genuine activity via GPS and machine learning, and **pays out automatically every Friday night** — one combined transfer per worker, aligned with how Swiggy and Zomato already pay their partners.

> **Core Insight:** Parametric insurance pays based on *objective data*, not subjective damage assessment. If rainfall crossed 80 mm/hr while Raju was working, he gets paid. The threshold is crossed — the payout fires. No adjuster, no form, no wait. And because payout is indexed to his actual predicted earnings, it is proportionally fair — not a flat Rs. 300 that means nothing to a Rs. 1,200/day earner.

2.1. Primary Persona — Raju, Swiggy Partner, Mumbai

Age	26	**Platform**	Swiggy
City / Zone	Mumbai, Bandra	**Vehicle**	Motorcycle
Avg. daily earnings	Rs. 600 – Rs. 900	**Peak hours**	12–2 PM, 7–10 PM
Primary risk	Monsoon flooding	**Payment**	UPI (PhonePe)

3. The Weekly Cycle — How GigShield Works

> **Design principle:** Swiggy and Zomato pay their delivery partners weekly. GigShield mirrors this cycle completely — predicting income weekly, accumulating disruption claims weekly, and disbursing one combined payout every **Friday at 23:59** — aligned with how workers already receive and budget their money.

Day & Time	System Event	What Happens
Saturday 00:00	**Week Resets + Prediction**	Fetch last 14 days from mock Swiggy/Zomato API for every active worker. XGBoost model predicts this week's earnings. Stored on WeeklyPolicy. Weekly accumulator reset to zero.
Mon – Fri	**Disruption Fires**	Weather threshold crossed during an active shift. AI pipeline runs: GPS fraud check → eligibility → income-indexed payout calculated. Claim saved as `approved`. Worker notified: "Claim of Rs. X approved. Paid Friday."
Mon – Fri	**Second Disruption**	Additional claim approved and added to the weekly accumulator. Worker can see running total in the app.
Friday 23:59	**Weekly Payout Batch**	All approved claims summed per worker. Plan weekly cap applied. **One UPI transfer per worker.** All claims marked `paid`. WebSocket push: "Your weekly payout of Rs. X is on its way!"
Saturday 00:00	**New Week Begins**	Accumulator reset. Friday payout flag reset. Saturday 00:00: new week opens instantly.

3.1. Three Scenarios

Scenario 1 — Monsoon Rain (Tuesday, Mumbai): Raju is on shift. Rainfall hits 85 mm/hr. GigShield detects the trigger, confirms Raju has 210 active minutes on shift, and calculates his claim against this week's XGBoost prediction (Rs. 3,800). Claim of Rs. 475 approved. Notification sent. Paid Friday.

Scenario 2 — Extreme Heat (Thursday, Delhi): Priya is on a Zomato shift. Temperature sustains 43°C for 2.5 hours. GigShield validates her GPS session and calculates her claim against her predicted weekly earnings. Claim approved, paid Friday with any other claims from the week.

Scenario 3 — Fraud Attempt (Blocked): A worker starts a shift but stays stationary at home. Heavy rain fires at 6:45 PM. The AI assigns a fraud score of 78/100 — above the 70 threshold. Payout blocked. Claim routed to admin review (not auto-rejected, to protect legitimate workers from GPS signal issues).

4. The Payout Formula

4.1. Why Income-Indexed?

A flat Rs. 300 per disruption is arbitrary. A worker earning Rs. 1,200/day loses 4× more income than one earning Rs. 300/day when a flood forces both off the road. GigShield calculates payout proportionally, using the worker's own predicted weekly earnings as the baseline.

4.2. Formula

$$\text{Payout per Disruption} = \underbrace{50\%}_{\substack{\text{income}\\\text{share}}} \times \underbrace{\frac{E_{\text{week}}}{D_{\text{work}}}}_{\substack{\text{daily}\\\text{slice}}} \times \underbrace{t_{\text{overlap}}}_{\substack{\text{hours}\\\text{disrupted}}} \times \underbrace{M}_{\substack{\text{severity}\\\text{multiplier}}}$$

E_{week}	=	XGBoost-predicted weekly earnings (set every Saturday 00:00)
D_{work}	=	6 work days (Saturday to Thursday, rest day Friday)
t_{overlap}	=	Hours the disruption overlapped the confirmed active shift
M	=	Severity multiplier: 0.6× – 1.5×
Cap		Total weekly claims ≤ plan weekly payout cap

4.3. Severity Multipliers

Disruption	Threshold	Multiplier	Severity
Moderate Rain	>40 mm/hr for 90 min	0.6×	MEDIUM
Heavy Rain	>80 mm/hr for 60 min	1.0×	HIGH
Extreme Heat	>42°C	1.0×	HIGH
Severe Pollution	AQI >300	1.0×	HIGH
Flood Alert	Official government alert	1.5×	CRITICAL
Curfew	Section 144 / civic shutdown	1.5×	CRITICAL

4.4. Worked Example — Raju's Week

	Tuesday — Heavy Rain	Thursday — Flood Alert
Predicted weekly earnings	Rs. 3,800	Rs. 3,800 (same)
Daily slice (÷ 6)	Rs. 633	Rs. 633
Hours disrupted	1.5 hrs	2.0 hrs
Severity multiplier	1.0×	1.5×
Income share	50%	50%
Claim amount	**Rs. 475**	**Rs. 950**
Status	Approved — wait Friday	Approved — wait Friday

Friday 23:59 — Combined	
Tuesday + Thursday total	Rs. 475 + Rs. 950 = Rs. 1,425
Standard plan weekly cap	Rs. 200
Final UPI transfer	**Rs. 200 (capped)**
Notification	"Weekly payout of Rs. 200 sent! Heavy rain + flood alert covered."

4.5. Key Design Decisions

- **Why 50%?** Full replacement creates moral hazard — workers stop working when it rains. 50% meaningfully reduces income shock while preserving the incentive to work. Actuarially, it keeps loss ratio within the 55–65% target.
- **Why 6 work days?** Most Swiggy/Zomato partners work Saturday–Friday cycle. Using 6 as the denominator gives a more accurate daily slice than 7.

- **Why store prediction on the policy?** Every disruption this week uses the same Saturday 00:00 baseline — no mid-week recalculation. Fully auditable by admins.
- **Why Friday 23:59?** The insurance week runs Saturday 00:00 to Friday 23:59 — one clean 7-day window. Payment on Friday night gives workers funds before the weekend. The new week opens at Saturday 00:00 immediately after, with zero gap in coverage. One combined transfer has higher psychological impact than multiple nightly credits and lower operational overhead.

5. XGBoost Earnings Predictor

5.1. Platform Earnings Connection

Every **Saturday at 00:00**, when the new insurance week opens, GigShield fetches **14 days of earnings history** from the Swiggy/Zomato partner API for each active worker. Each day's record contains: net earnings, gross earnings, orders completed, active hours, peak hours worked, day of week, and whether the day was disrupted.

> **Why two weeks, not one?** One week is too noisy — a single bad week (illness, festival) skews the prediction. Two weeks captures weekday/weekend patterns, disrupted vs. clean days, and genuine earning trends. More than four weeks over-smooths recent changes in work patterns.

5.2. The Model

A personalised XGBoost regression model is trained on each worker's own 14-day history. Each worker gets their own model — not a global average. Shallow trees (`max_depth=3`) prevent overfitting on short windows.

Feature	Description	Why It Matters
Day of week	0 (Mon) – 6 (Sun)	Weekend orders 20–30% higher
Is weekend	Binary 0/1	Friday–Sunday premium
7-day rolling avg	Mean, last 7 clean days	Recent trend baseline
3-day rolling avg	Mean, last 3 days	Short-term momentum
14-day rolling avg	Mean, full history	Long-term baseline
Earnings std dev	Volatility	Prediction confidence
Avg orders (7d)	Order volume trend	Platform demand signal
Peak hours worked	12–2 PM, 7–10 PM	Worker availability
Rain forecast (mm)	OpenMeteo 7-day	Weather earnings impact
AQI forecast	IQAir AirVisual	Pollution impact
Zone flood risk	0.0 – 1.0	Historical disruption rate
Earnings trend	Linear slope of history	Rising or falling income

> **Fallback:** Fewer than 5 days of history (new workers) triggers a **weighted moving average** — 55% weight to the last 3 days, 20% to the 3 days before that, 10% to older data, with a 1.15× weekend multiplier. Graceful degradation built in.

6. Weekly Plans & Dynamic Premium

6.1. Base Plans

Plan	Premium	Weekly Cap	Disruptions	Cycle
Basic	Rs. 20/wk	Rs. 150	1	Sat 00:00 – Fri 23:59
Standard	Rs. 30/wk	Rs. 200	2	Sat 00:00 – Fri 23:59
Premium	Rs. 50/wk	Rs. 300	3	Sat 00:00 – Fri 23:59

6.2. Dynamic Premium Formula

$$P_{\text{weekly}} = P_{\text{base}} \times \underbrace{Z}_{\substack{\text{zone}\\\text{risk}}} \times \underbrace{S}_{\substack{\text{seasonal}\\\text{factor}}} - \underbrace{L}_{\substack{\text{loyalty}\\\text{discount}}}$$

Zone Risk (Z)	Historical disruption rate in GPS zone	1.0 – 1.3
Seasonal Factor (S)	7-day forecast disruption probability	1.0 – 1.2
Loyalty Discount (L)	Clean claims history reward, 8+ weeks	Rs. 0 – Rs. 3

Example: Standard (Rs. 30) × 1.2 (Bandra zone) × 1.1 (monsoon) − Rs. 3 (loyalty) = **Rs. 36.60/week**

6.3. Insurance Pool Sustainability

10,000 workers × Rs. 30/wk	Rs. 3,00,000 weekly premium pool
Expected disruption rate	10% of worker-weeks
Expected weekly payouts	Rs. 60,000 – Rs. 90,000
Target loss ratio	55% – 65% (actuarially sustainable)
Catastrophic maximum liability	10,000 × Rs. 300 = Rs. 30,00,000 (within 10-week reserve)

7. Parametric Triggers

A payout fires only when **both** conditions are true: **Condition A** — disruption threshold met; and **Condition B** — disruption overlapped worker's confirmed active GPS session by ≥ 30 minutes.

#	Trigger	Threshold / Condition	Source	Multi.
1	Heavy Rainfall	>80 mm/hr for ≥60 min	OpenWeatherMap	1.0×
2	Extreme Heat	>42°C for ≥2 hr	OpenWeatherMap	1.0×
3	Severe AQI	AQI >300 for ≥3 hr	IQAir AirVisual	1.0×
4	Urban Flooding	Flood alert + >150 mm/24 hr	OWM + GPS	1.5×
5	Curfew / Sec.144	Movement restriction in zone	Govt. alert feed	1.5×
6	Short Shift	Active time <4 hr + weather event	GPS + Weather	1.0×

Trigger 6 catches weather-forced early exits: even if no single threshold was breached, a shift

> cut short by deteriorating conditions qualifies. **All triggers cover income loss only.** Vehicle damage, health, and accidents are excluded per platform rules.

8. AI & Machine Learning Architecture

AI fires at four points in the workflow:

When	Model	What It Does
Registration	Dynamic Premium Engine	Computes risk score from zone, city, vehicle, season → personalised premium
Saturday 00:00	XGBoost Predictor	Fetches 14-day platform history → predicts weekly earnings → stored on policy
During shift	Activity Classifier	GPS speed, stops, distance, cluster density → ACTIVE / INACTIVE / REVIEW
At disruption	Fraud Detection Engine	GPS spoofing, impossible speed, low activity, bot patterns → fraud score 0–100

8.1. Fraud Detection Rules

Flag	Trigger Condition	Score
GPS Spoofing	>80% GPS points with 0 m accuracy	+35
Impossible Speed	>120 km/h between consecutive points	+20/event
Low Activity	<15% active movement during shift	+25
Bot Pattern	>90% cardinal heading movement	+25
Low Coverage	<10 GPS points recorded	+30
0 – 39	Auto-Approved: pay immediately	
40 – 69	Manual Review: admin checks before Friday payout	
70 – 100	Auto-Rejected: blocked; fraud logged	

8.2. Dynamic Premium Risk Score Components

Factor	Weight	Example
Disruption history in zone	30%	Bandra: 67 events → HIGH
Zone flood risk	25%	Bandra: 0.85
Seasonal factor	20%	Monsoon season: 85/100
Vehicle type	15%	Bicycle: 80 (most exposed)
Platform	10%	Zepto: 65 (faster SLA pressure)

9. Eligibility Rules

All six conditions must be true for a claim to be approved:

	Rule	Detail
1	Active weekly policy	Worker has paid this week's premium
2	Min 3 active shift hours	GPS confirms ≥180 min genuine movement
3	Disruption overlap 30+ min	Disruption window overlaps shift by ≥30 min
4	Fraud score < 70	AI GPS classifier clears the shift
5	Weekly limit not reached	Below plan's max disruptions per week
6	Weekly cap not exhausted	Accumulated payout < plan weekly cap

> **Scores 40–69 go to manual review, not auto-rejection.** This protects legitimate workers from false positives caused by GPS signal loss, indoor delivery stops, or poor network areas. Admin can approve before the Friday payout runs.

10. Technology Stack

Backend — Java 17 + Spring Boot 3.2

- REST API with JWT authentication
- Spring Security + OTP via Fast2SMS
- WebSocket / STOMP for real-time push
- `@Scheduled` Saturday 00:00 (reset + predict) + Friday 23:59 (payout)
- `@Async` claim processing pipeline
- H2 (dev) / PostgreSQL (production)

AI Service — Python 3.11 + FastAPI

- `POST /api/platform/earnings` — mock platform API
- `POST /api/predict-earnings` — XGBoost
- `POST /api/check-eligibility` — payout calc
- `POST /api/analyze-gps` — fraud detection
- `POST /api/risk-score` — premium engine
- `POST /api/friday-batch` — weekly payout

Frontend — Mobile-First Web (HTML/CSS/JS)

- `login.html` — worker + admin split login
- `register.html` — 3-step onboarding
- `dashboard.html` — live shift + GPS
- `subscribe.html` — AI-priced plans
- `claims.html` — weekly payout history
- `admin.html` — analytics + simulator

External APIs (all free tier)

- OpenWeatherMap — rain + temperature
- Open-Meteo — 7-day forecast (**no key needed**)
- IQAir AirVisual — real-time AQI
- Razorpay — UPI payout simulation
- Fast2SMS — OTP delivery

> **Why mobile-first web, not a native app:** No App Store barrier; works on budget Android on 3G; single codebase for mobile workers and desktop admin; faster hackathon iteration. Judges can open it in any browser with a single link.

11. System Architecture

```
Frontend (HTML/CSS/JS) -- GitHub Pages
↓ REST API + JWT
Spring Boot Backend -- Railway.app
Java 17 · Spring Security · WebSocket · H2/PostgreSQL
↓ HTTP                          ↓ WebSocket
Python FastAPI                  Worker Browser
XGBoost · Fraud ML              Real-time push notifications
Render.com                      "Rs. 200 payout Friday!"
↓ HTTP (all free tier)
OpenWeatherMap · Open-Meteo · IQAir AirVisual
Mock Swiggy/Zomato Earnings API
```

11.1. Weekly Cycle — Saturday to Friday

1. **Saturday 00:00** — `@Scheduled(cron = "0 0 0 * * SAT")` fires. Accumulator reset. XGBoost fetches 14 days of platform earnings and predicts this week's income. Stored on the worker's active `WeeklyPolicy`.
2. **Saturday – Friday (any time)** — Disruption detected during active shift. GPS fraud check → eligibility check → payout calculated. Claim saved as `approved`. Worker notified instantly. Claim accumulates.
3. **Friday 23:59** — `@Scheduled(cron = "0 59 23 * * FRI")` fires.
4. Collect all `approved` claims created since Saturday 00:00.
5. Group by worker. Sum all claim amounts per worker.
6. Apply plan weekly cap.
7. One UPI transfer per worker via Razorpay.
8. Mark all claims `paid`. Reset accumulator.
9. WebSocket push: "Rs. X weekly payout sent. New week starts now."
10. **Saturday 00:00** — Next insurance week opens immediately. Continuous coverage. No gap between weeks.

12. Covered Zones

Zone	City	Flood Risk	Historical Disruptions
Bandra	Mumbai	85%	67 events
Andheri	Mumbai	75%	58 events
Salt Lake	Kolkata	70%	52 events
T Nagar	Chennai	60%	45 events
Koramangala	Bangalore	45%	28 events
Sector 18	Noida	40%	19 events
Connaught Place	Delhi	35%	22 events
Jubilee Hills	Hyderabad	30%	15 events

13. Development Roadmap

Phase	Dates	Key Deliverables
Phase 1 Ideation & Foundation	Mar 4 – 20	Problem research, persona, architecture, DB schema, this document, 2-minute strategy video
Phase 2 Automation & Protection	Mar 21 – Apr 4	Worker registration, GPS shift tracking, XGBoost predictor, dynamic premium, 6 parametric triggers, weekly payout pipeline, worker dashboard
Phase 3 Scale & Optimise	Apr 5 – 17	Advanced fraud detection, Razorpay integration, full admin analytics, disruption simulator, pitch deck, 5-min demo video

14. Worker Experience — End to End

1. **Saturday 00:00** — New insurance week opens automatically. GigShield runs XGBoost prediction in the background. Raju opens the app and sees: *"New week started. Your predicted earnings this week: Rs. 3,800. You are covered up to Rs. 200 if disruptions hit (Saturday to Friday)."*

2. **Tuesday afternoon** — Rain disruption fires. Instant notification: *"Heavy rain detected in Bandra. Your shift was active. Claim of Rs. 475 approved. Will be paid Friday 23:59 with your weekly summary."*

3. **Thursday evening** — Flood alert fires. Instant notification: *"Flood alert in Bandra. Claim of Rs. 950 approved. Your week total: Rs. 1,425 — capped at Rs. 200 (Standard plan). Paid Friday 23:59."*

4. **Friday 23:59** — Payment arrives. *"Your weekly protection payout of Rs. 200 is on its way! Covered: heavy rain + flood alert (2 disruptions this week). Credited to your UPI."*

5. **Saturday 00:00** — New week opens immediately. *"New insurance week started. Coverage active. Predicted earnings: Rs. 3,920."* Zero gap in coverage.

15. Team

Name	Role
Saheli Roy	Backend Development
Krishnendu Malick	AI & Backend
Rishav Kumar	Frontend & UI/UX
Aniket Das	Frontend Development
Rishika Singhadeo	UI/UX Design